

An introduction to eCommerce

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eCommerce is set to revolutionise the way we do business. The companion papers to be found in the themed section of this edition outline some of the major technologies that underpin eCommerce. This paper sets the scene.

1. What is eCommerce?

Electronic commerce — or eCommerce — is defined many ways, and the definition is somewhat arbitrary. Perhaps the most useful way of describing eCommerce is to use it to link it to trading:

‘eCommerce is trading by means of new communication technology. It includes all aspects of trading, including commercial market making, ordering, supply chain management, and the transfer of money.’

‘New communication technology’ in the definition means everything beyond voice telephony, fax and telex, which have been with us for many years. It is the new technologies of the Internet, together with the prodigious advances in IT generally, that have enabled revolutionary changes in the way we trade — eCommerce.

This definition is used throughout the accompanying papers in the themed section of this issue.

2. Why is eCommerce happening?

As mentioned above, there have been rapid and continuing developments in IT. PCs are cheap and powerful, with increasingly rich and user-friendly interfaces; they are tools for everyone, not just experts. Microprocessor and communications technology is cheap enough that almost any piece of equipment may be intelligent — and connected; and it is ‘connected’ that provides the key to eCommerce. Trading is about communication — between buyer and seller, provider and consumer, manufacturer and user, retailer and wholesaler.

Specialist communication has been available to provide some eCommerce for many years. Electronic data interchange (EDI) has linked large companies with established trading relationships (and the money to implement the technology). BACS and CHAPS provide inter-bank clearing, and the major credit card companies

provide central card clearing. However, the costs of entry were high, limiting participation mainly to large or specialist companies. A mass market was out of the question.

The key to the eCommerce revolution — unsurprisingly — is the Internet. Not because the Internet does anything previous networks did not do, nor because it is better. Simply because it has become **the** network — the ubiquitous connectivity for virtually all non-voice communication. The result is that the cost of connecting and communicating has tumbled; and because it is the universal network, you can be sure that all trading partners will use it too. The flat rate ‘all-you-can-eat’ cost structure has greatly encouraged global thinking. No one on the Web is more than a click away. The entry costs for the Internet are low enough to encourage small to medium sized enterprises (SMEs) and private consumers to use it. There were fewer than 40 million people connected to the Web in 1996, 100 million by the start of 1998, and over 150 million now [1]. This latter figure is around half the number of telephone connections worldwide. For several years Internet traffic has been increasing by 8 times every year. Even more significantly — Internet users will be transacting business worth around £184bn (\$300bn) by 2000 [2], possibly more according to other research organisations.

There are other enabling factors that encourage electronic trading, such as the increasing use of non-cash methods of payment (such as credit card) that do not require physical presence. Multinational companies are looking for even more globalisation of trading. With fast and low-cost communication to potential customers, the attraction of trading electronically via the Web is obvious. However, the nature of trading by the Web brings other new advantages:

- the Web is targeted, narrowcast rather than broadcast — it is easier (and potentially cheaper) to identify the right customers, and identify their needs,

- the Web is two-way — much faster interaction with customers is possible than by TV or newspaper advertisements, or by mail-shots,
- the Web is global — no other trading method offers an instant global market,
- the Web is cheap — allowing lower cost of entry than many conventional methods of trading.

Clearly there are also limitations in eCommerce. Where a customer buys goods electronically (rather than visiting a shop) it may not be possible to physically see or handle the item before buying. Such goods must also be physically delivered. Some service industries do not, on the face of it, lend themselves to eCommerce. You cannot deliver a pizza electronically! However, you can order it by phone, and there are cost savings in taking the order (and paying for it) electronically.

3. The eCommerce process

There is plenty of evidence of eCommerce everywhere one looks on the World Wide Web. Many commercial Web sites have catalogues, and quite often it is also possible to place orders on the Web — but eCommerce is more than this. It includes everything from sourcing to settlement, and all the processes that underlie trading, as shown in Fig 1.

There are many types of trading — retail sales to consumers, wholesale trading, business-to-business trading, etc. There are also many types of market needed. For

example, mass-produced items might be detailed in a catalogue with listed prices, but unique items may be sold in an auction where the price is determined interactively between potential buyers and the seller.

As a result eCommerce is also characterised by enormous diversity of requirement and implementation. Each of the four groupings of functions in Fig 1 can be implemented in many different ways. For example, 'sourcing' may be implemented by Web catalogues, by specialised quotation systems, by order-matching systems, by ITT, by auction, or by numerous other ways. 'Order placing' may be by Web forms, by e-mail, by EDI, by telephone sales agent, etc.

Few eCommerce products tackle more than part of the problem. For example, most catalogue systems concentrate on 'sourcing' processes, and sometimes 'order placing'. Such systems are often called 'trading environments'. However, there are support elements that are common to many aspects of eCommerce. This can be seen from Fig 2.

Elements such as trust (including integrity and security) are common to all aspects of eCommerce.

The remainder of this paper considers one common subsystem of eCommerce — the trading environment (roughly sourcing and order placing) and some of its support elements — security, trust and payments. These and other eCommerce issues are expanded upon in the associated papers in this issue.

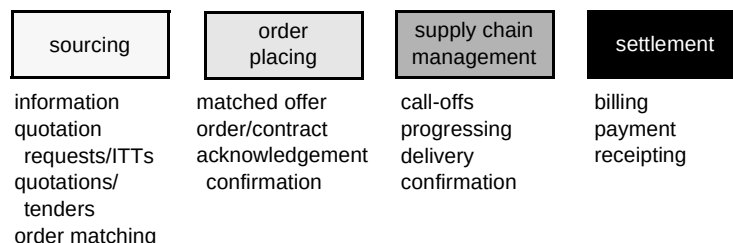


Fig 1 The trading processes.

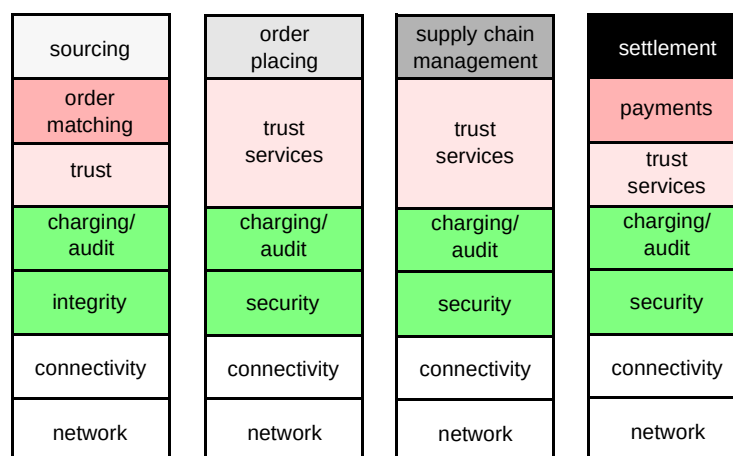


Fig 2 Commonality of support elements.

4. Electronic trading environments

4.1 Electronic data interchange

The eCommerce concept is not new — businesses have been carrying out commercial transactions using EDI for many years [3, 4]. EDI enables businesses to exchange structured messages (orders, invoices, etc) between computer systems without the need for human intervention. The messages are usually carried over proprietary value added networks (VANs), such as BT's EDI*Net, that provide store-and-collect mail services, audit trails and protocol resolution facilities for their subscribers.

The EDI business model is based on established trading agreements between buyers and sellers. It operates very effectively in 'hub-and-spoke' configurations where a dominant buyer (the hub) dictates the terms of relationships with a large number of smaller suppliers (the spokes). The model offers major strategic and operational benefits to the corporate hubs that need to centralise their operations and reduce the manual re-keying of transaction data [4]. However, the benefits to the spokes, typically SMEs, are less clear-cut — EDI has a reputation for being costly to operate and the SME may have to subscribe to a number of mutually incompatible VANs in order to satisfy the requirements of different trading hubs.

The advent of the Internet has created new opportunities for EDI, providing a ubiquitous, open network that could potentially replace the VANs or extend their reach into new markets [4]. Current initiatives include both Web front-ends on to existing VANS (e.g. GE Tradeweb) and full end-to-end solutions for the Internet (e.g. Harbinger's TrustedLink Templar). Standards such as SimplEDI, EDI/Internet and XML/EDI are emerging to address the Internet EDI market [5].

4.2 Internet trading

The Internet offers new opportunities for commerce that go well beyond simply replicating EDI [6]. Perhaps the most obvious example is the emergence of direct on-line selling to consumers. Unlike EDI, direct selling relies on the generation of spontaneous trade between parties that have no formal trading agreement. The business model is similar to mail order, with suppliers publishing a catalogue of products on a Web site and seeking to attract customers to the site by strong promotion of their brand. This model has attracted considerable attention in recent years, stimulating a rush of businesses to get on-line. However, even the most successful on-line stores (e.g. Amazon.com) have struggled to become profitable and businesses are increasingly seeing their Internet stores as a strategic investment, rather than an immediate money-spinner.

Superficially, direct selling offers the prospect of large-scale disintermediation, eliminating middle-men from supply chains, and it is certainly true that some traditional intermediary roles will become threatened as businesses and

individuals re-evaluate their trading patterns in response to new eCommerce opportunities. However, far from eliminating the need for intermediaries, the Internet is currently generating a wide range of new opportunities in the facilitation and mediation of electronic trading environments. Portal sites (e.g. Yahoo!, Excite), on-line auction sites (e.g. eBay) and electronic brokers (e.g. Auto-by-tel) have provided some of the major success stories in the early stages of the eCommerce revolution.

In the business-to-business market there is currently considerable interest in the development of Internet-based procurement systems for indirect goods (otherwise known as MRO — maintenance, repair and operations), such as stationery and IT equipment [6]. Ariba, CommerceOne and Infobank are all developing propositions in this area. These systems provide a customised buyer catalogue for each purchasing organisation, generated from the relevant contents of participating supplier catalogues. The buyer catalogue can be made available over the company intranet, providing employees with up-to-date product information whenever they need it.

4.3 Intelligent on-line purchasing

On-line trading environments are still very much in their infancy. Over the next few years we can expect to see a rapid increase in the functionality of these environments that will assist both buyers and sellers [7]. Intelligent purchasing advisors will assist buyers in specifying their product requirements, searching for product information and selecting the best supplier. This will inevitably lead to a shift in the balance of power from sellers to buyers [6, 8]. Sellers, on the other hand, will benefit from enhanced data-mining and user-profiling techniques, providing new opportunities for targeted marketing.

Intelligent agents are likely to play a key role in these advanced trading environments. Pioneering work on the MATE project at BT Laboratories [9] has demonstrated the use of intelligent agents to represent buyers, sellers and virtual market providers in a fully distributed environment. MATE enables comparison shopping based on the buyer's preferences across a range of product and supplier attributes.

5. Security, trust and payments

5.1 Security issues

Concerns over the security of the Internet, much hyped by the media, have done little to reassure users that the Internet is a safe place to trade. These concerns must clearly be addressed if buyers and sellers are to have the confidence to trade electronically. Major security issues include:

- confidentiality — data must not be visible to eavesdroppers,
- authentication — communicating parties must be certain of each other's identity and/or credentials,

- integrity — communicating parties must know when data has been tampered with,
- non-repudiation — it must be possible to prove that a transaction has taken place.

Much of the public concern centres on confidentiality — is it safe to send personal information (such as credit card details) over the Internet? However, the other issues are probably more important to eCommerce — it is the issues of trust that are essential for trading.

5.2 Cryptography

Cryptography [10] is the major technology used to address these security issues. Two forms of cryptography are in common use — symmetric encryption and asymmetric encryption (also known as ‘public key cryptography’). Both use cryptographic algorithms and keys to encode and decode data, where ‘keys’ are the parameters used in the mathematical encryption process. The strength of the encryption increases exponentially with key length.

In symmetric key systems, the sender and receiver must use identical keys. The security of these systems depends on protecting the secrecy of the keys, therefore key distribution requires a separate, secure distribution channel — a disadvantage as far as widespread use in eCommerce is concerned. However, as symmetric encryption algorithms are much faster than public key algorithms, they are preferred for the encryption of large data files. The Digital Encryption Standard (DES) is an example of a symmetric encryption system widely used for communications.

Public key schemes use pairs of related keys. Each user has a private key, which is kept secret, and a public key that is published and made readily available to others. Typically, a document encrypted with either of the keys may be decrypted using its counterpart. The key pair may be used to provide:

- confidentiality — the public key encrypts a document so that it can only be read by the owner of the corresponding private key, this approach often being used as a secure way of exchanging symmetric keys,
- digital signatures — documents signed used the signer’s private key may be validated using the corresponding public key, thus providing a basis for authentication, integrity and non-repudiation.

The most well-known example of a public key cryptosystem is RSA, named after its creators Rivest, Shamir and Adleman.

5.3 Public key infrastructures

The overall security of public key cryptography schemes is crucially dependent on the implementation and management of an appropriate public key infrastructure (PKI). PKIs provide the facilities required to manage the

life cycles of public key pairs, including user registration, key pair generation, certification, key recovery and certificate revocation. Although PKIs have existed for more than 20 years, the advent of eCommerce has for the first time provided a strong business case for the roll-out of PKIs to the mass market. A range of new business opportunities has emerged, particularly in the provision of trusted third party (TTP) services [11] and PKI solutions. The scaling-up of solutions to address the mass market has started to expose some of the limitations of the technology.

BT has recently entered the eCommerce TTP market with the launch of its certification service, Trustwise [12]. It was recognised in 1997 that there was a limited window of opportunity for BT to set up as a public Certification Authority in the UK — the digital certificate market was maturing rapidly and there were large benefits to early market entry. BT therefore decided to partner with Verisign, a leader in the US certification market, to speed entry to market and user acceptance. The initial Trustwise service has been delivered in three phases, commencing July 1998.

In the longer term, BT plans to develop its Trustwise product line to encompass a wider range of services that facilitate trust in eCommerce [13]. This is likely to include high value certification [14] — certificates that go beyond just proving a user’s identity, to providing evidence of character and ability. High value certificates are stretching the requirements for PKI technology still further, stimulating new research in areas such as key management, certificate revocation/control, liability management and trust infrastructures.

5.4 Smart cards

Currently, many users of public key cryptography schemes store private keys in software (e.g. using encrypted directories on their hard disks). This is far from ideal — the private key is only protected by a single password and there is little support for mobility.

Smart cards [15] provide a convenient, mobile alternative for key storage. Security of the system is increased, since access to the private key is now controlled by something the user owns, as well as something the user knows (the password). The similarities between smart cards and existing credit/debit cards also ensure that many users are already familiar with the basic principles of card usage and management. Current barriers to widespread adoption are largely associated with the lack of agreed smart-card standards (e.g. Microsoft and Netscape support different standards for integrating smart cards with their Web browsers) and the corresponding lack of smart-card ‘slots’ in PCs.

Smart cards offer the potential for far more than just key storage. With recent developments in smart-card technology, it is now possible to generate a digital signature (or decrypt a session key) on board the smart card. This

ensures that the private key can remain within the secure environment of the smart card at all times, rather than having to enter the memory of the computer for processing. Some cards are also starting to carry out on-board generation of key pairs, enabling the whole life cycle of the private key to be managed within the secure environment of the card. Such developments will be essential if evidence based on use of digital signatures is to be strong enough to stand up in a court of law.

Key management is by no means the only use for smart cards within eCommerce. Smart cards are increasingly being used to support a wide range of applications, including user authentication [15], electronic cash systems [15, 16] and loyalty schemes. With the advent of multi-application smart cards, it is expected that several of these functions will be combined on to a single card. For example, financial services companies may issue a single smart card combining traditional credit/debit card facilities with an electronic purse, strong authentication for home banking and key management to support secure electronic payments.

5.5 *Electronic payments*

Electronic payments [16] provide the ultimate test of security and trust in eCommerce. By far the most popular mechanism for payment on the Internet today is the credit card. Card details are transported across the Internet using public key-based protocols such as SSL and SET [17]. The existing international credit card infrastructure is exploited to provide instant on-line payment for a wide range of different currencies. The only major limitation is the high transaction cost involved, making it uneconomic to use credit cards to collect payments below about £5.

There is still a lot of debate as to whether users will be prepared to pay for low-value items (e.g. pages of information, music and video clips) delivered over the Internet, or whether advertising will remain the dominant mechanism for funding commercial Web sites. Nevertheless, a lot of work has been carried out in recent years to develop 'micropayment' systems capable of collecting small payments below the value at which a credit card transaction becomes cost effective. A wide range of systems has been developed, including both 'software wallets' (e.g. Cybercash, KLELine, Cybank) and server-based systems (e.g. OpenMarket, BTArray, WorldPay). However, none has become a dominant standard. It is now looking increasingly likely that smart card-based electronic purse schemes (e.g. Proton, Visa Cash, Mondex), backed by the major credit card companies, will ultimately win the battle for the micropayments market. Smart cards provide a secure, mobile environment for the storage of monetary value that can be used both in the physical world and on-line. The uptake of electronic purse schemes is currently very slow, with no short-term business case for usage in High Street transactions [15]. However, the market could

suddenly explode as standardised purse schemes and smart card interfaces to PCs come on to the market.

6. **Legal and regulatory issues**

Many of the biggest issues in eCommerce today have very little to do with technology. Rather, they relate to the external environment within which eCommerce operates. The Internet does not recognise administrative and legislative boundaries and, as such, eCommerce falls outside the well-understood legal and regulatory frameworks within which trading has traditionally taken place. The issue is not so much an absence of legislation, but rather confusion over which of the many possible legal systems should apply. Taxation presents similar problems.

As a major global communications provider, BT has an important role to play in tracking, and where appropriate influencing, the activities of external bodies such as the UK Government, the European Commission and international trade organisations. Issues attracting attention at present include the legal recognition of electronic signatures and the nature of electronic contracts [5]. UK legislation on electronic signatures and the licensing of TTPs is proposed as part of the eCommerce Bill to be debated in the current session of Parliament (1999).

7. **BT eCommerce trials**

In a new and fast-moving area like eCommerce, there is no real substitute for practical, hands-on experience to gain a detailed understanding of the key issues. Over the past few years, BT has run a range of trials aimed at exploring different aspects of eCommerce. These have included:

- electronic retailing [18],
- business-to-business trading (including back-end integration) [19],
- implementation of eCommerce within a community of SMEs [20].

The SME market is considered to be a key target area for eCommerce as we move into the next millennium.

BT Laboratories runs an eCommerce test bed that facilitates the rapid development and operation of eCommerce trials [18]. The platform supports a growing portfolio of eCommerce components, enabling new services to be rapidly configured through component reuse.

8. **Future directions**

The eCommerce revolution is only just beginning! Over the next few years e-Commerce can be expected to have a profound influence on the way every individual and business conducts trade [8]. The advent of interactive digital TV will take eCommerce beyond the realms of the PC-literate to create a truly mass-market phenomenon.

Screenphones, personal digital assistants, kiosks and a host of other 'information appliances' will provide access to eCommerce services 'anytime, anyplace, anywhere' [21]. New business models will emerge, creating new threats and new opportunities. The trend towards globalisation will increase, with its associated challenges for taxation, law and regulation. There will be winners and there will be losers. Only one thing is certain — life will never be quite the same again!

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Peter Skevington manages the electronic commerce research group at BT Laboratories. His group is involved in a wide range of activities associated with the future of eCommerce, including work on trust services, trading environments, intelligent agents and on-line shopping applications. He has led BT's trust services research project for the past three years and carried out early work on the creation of BT's digital certification service, Trustwise. He has an MA in Physics from Cambridge University and a PhD in optoelectronics from Cardiff University.